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Leveraging facial landmarks improves generalization ability for deepfake detection

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ABSTRACT

Recently, facial forgery technology has become increasingly sophisticated and published datasets aim to cover a wide range of data variations. Existing deepfake detection models have benefited from the powerful feature embedding of deep networks and carefully designed fine-tuning modules, resulting in an excellent performance on in-dataset evaluations. However, the performance declines in cross-dataset evaluations due to various forgery methods and dataset shifts. In this study, we concentrate on the generalization issue of deepfake detection and find that forgery traces appear to gather around the facial interest points even manipulated by different forgery methods. To facilitate this, we propose a Trail Tracing Network (TTNet) to capture the generalized feature representation, which leverages facial landmarks to eliminate redundant information and expand the forged traces in the feature space. We conduct extensive experiments on the widely employed benchmarks, including FaceForensics++, DFDCp, and Celeb-DF. Experimental results demonstrate the outstanding generalization ability of our method against existing state-of-the-art methods by a large margin. In addition, the proposed method also exhibits excellent performance on the in-dataset evaluation.

1. Introduction

With the rapid advancement of image synthetic technologies based on Generative Adversarial Networks (GANs) [1], an extensive array of highly realistic forged images and videos have been synthesized. Nevertheless, the malicious use of deepfake technology, such as fabricating political speeches, creating counterfeit celebrity statements, and disseminating false information, has significantly heightened public concern and eroded societal trust [2]. Moreover, due to the easy accessibility of deepfake techniques, ordinary individuals may also become targets of such attacks. Given these implications, the design of a reliable deepfake detection system has become increasingly important.

Currently, the performance of facial forgery detection techniques is mainly examined by in-dataset evaluations, i.e., the training and test sets are from the same distribution of facial forgery data. During the training phase, the model learns the specific traces of the manipulated patterns and fits all the characteristics of the data as much as possible. Subsequently, during the testing phase, the model fits various traces discovered in the training phase according to the features extracted from the image and determines if the image is forged. However, there is a fatal drawback that if the model encounters a completely new forgery pattern or large data shifts between the training set and the test set, the features extracted by the model may not be able to find a reasonable basis for forgery detection, leading to a significant decrease in the model's detection performance [3,4]. Our work aims to address the generalization issue for deepfake detection, wherein the core is to recognize and enhance the characteristics of forged traces that are commonly produced by various forgery methods.

Fig. 1 investigates the trails left by different forgery methods. We can observe that forgery traces usually appear around facial features and contours. This is because most facial forgery methods merge two faces by placing the facial features of one face on another, inevitably leading to anomalies at the fusion boundary. To further explore this issue, we conduct a survey of forgery regions generated by four different facial forgery methods in the FaceForensics++ (FF++) [5] dataset. The results are shown in Fig. 2, where spatial locations of forgeries are summarized into five clusters, including eyes, nose, mouth, facial contour, and others. It is clear that most of the forgeries appear around facial features and contours and the probability of forgeries appearing in other spatial locations is low across all types of forgery methods. Based on this, we argue that deepfake detectors should focus on these regions prone to forgeries rather than searching for potential clues throughout the entire face in a random manner. To this end, we propose to leverage facial landmarks in deepfake detectors. Facial landmarks contain rich spatial information related to facial semantics, which is beneficial to the generalizable representations in deepfake detection.

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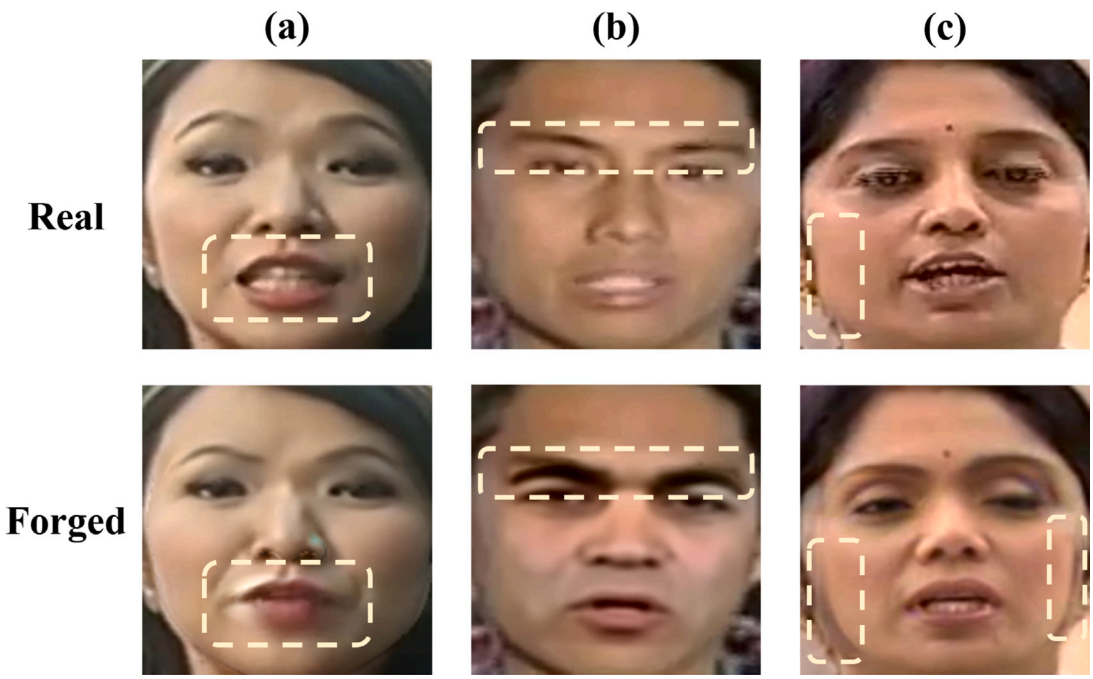

Fig. 1. Traces left by different manipulated methods. (a) Face2Face: The coloration around the mouth of the manipulated face appears anomalous. (b) Deepfakes: The central eyebrow region shows clear evidence of incomplete coverage. (c) FaceShifter: The cheek area displays fusion deformation. These traces of forgery all manifest in the five senses and facial contours even manipulated by different forgery methods. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

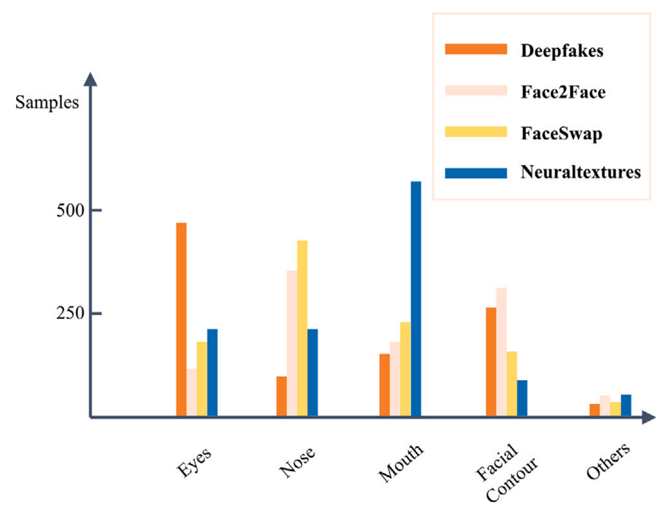

Fig. 2. In the FF++ dataset, forgery regions are categorized and statistically analyzed based on the distribution of facial landmarks. Forgery regions can be broadly grouped into five categories, including eyes, nose, mouth, and facial contour. Therefore, to capture common forgery traces left by different forgery methods, face forgery detectors should focus on these areas.

To leverage the facial landmarks, we propose a Trail Tracing Network (TTNet) to perceive the common forged traces in various forgery methods. The network includes a Trace Enhancement Module (TEM) that utilizes facial landmark features to reinforce the forged traces. This enhances the generalization ability of the proposed model for the deep forgery task. In summary, our contributions are three-fold:

- We investigate the forgery traces by different facial forgery methods in the FaceForensics++ dataset and observe that most of the forgery traces are distributed on the facial landmarks even with different manipulated methods. Based on this observation, to improve the generalization ability, we propose to leverage facial landmarks in deepfake detection models.
- We propose a Trail Tracing Network (TTNet) that identifies forgery traces guided by facial landmarks. It consists of two modules, including a Generalization Enhancement Module (GEM) and a Redundancy Elimination Module (REM). The former enhances the forgery traces and the latter eliminates redundant information.
- We conduct extensive experiments including both in-dataset experiments and cross-dataset experiments. Experimental results demonstrate the detection performance as well as excellent generalization ability against numerous state-of-the-art methods.

2. Related work

2.1. Deepfake detection

In this section, we summarize recent works on face forgery detection, focusing on two categories of work: those aimed at improving detection performance and those aimed at improving generalization performance.

Detection performance. Many researchers have manually designed features for the classifier input based on the distribution characteristics of face forgery detection datasets. For example, Yang et al. [6] used head pose errors to detect the authenticity of fake faces, while Peng et al. [7] detected forgery traces in facial videos by analyzing the distribution of eye gaze. With the rapid development of deep networks, some researchers have introduced CNNs into the task of face forgery detection, such as Cozzolino et al. [8] and Bayar et al. [9], all of whom used simple CNNs to achieve excellent detection results on a single dataset. Later, RNNs were used to incorporate the temporal dimension of video data into the judgment of fake videos. Masi et al. [10] used a two-stream RNN network to distinguish the authenticity of data based on abnormal facial movements. In addition to the aforementioned work, several studies have focused on enhancing the robustness of forgery detection [11]. For instance, Liao et al. [12] proposed a method that analyzes facial muscle movement features to detect compressed Deepfake videos. Furthermore, beyond traditional forgery detection techniques, some researchers in recent years have attempted to infer the sequence of forgery operations by analyzing the localization of the forgeries [13,14]. These methods can achieve excellent detection performance when targeting specific datasets. However, due to their overly customized feature extraction and the redundancy of feature space in the network, their detection ability is significantly dropped when facing new forgery methods. Thereupon, improving the generalization of the model has become another challenge in the task of face forgery detection.

Generalization performance. As forgery techniques develop, existing detection methods struggle with unknown forgeries. Enhancing detector generalization and cross-dataset performance is crucial, and researchers are focusing on identifying common forgery traces across different generation methods. For example, Li et al. [15] and Yang et al. [16] found that forgery traces appear at the boundaries of synthesized regions in fake faces, and focused on detecting these traces to improve generalization. In addition to using color textures, Liu et al. [17] used DFT to extract robust features from the high-frequency components of fake images, achieving good generalization across datasets with a shallow neural network. Liu et al. [18] also used frequency domain modalities, proposing a novel attention consistency refined masked frequency forgery representation model algorithm toward generalizing face forgery detection. Wu et al. [19] combined frequency domain and RGB modalities with temporal features to obtain more robust and generalized representations. The use of cross-modality information is not limited to this, Liu et al. [20] proposed a hierarchical forgery classifier to improve generalization in visible light and near-infrared image detection. The model combines mixed-domain features from both image and frequency domains for better feature discrimination. Moreover, some researchers have extracted features with generalization based on the temporal distribution of video data. For example, Wang et al. [21] proposed a discrete cosine transform-based forgery clue augmentation network based on discrete cosine transform, achieving a more comprehensive spatio-temporal feature representation. The mentioned methods expand data information to improve generalization across datasets. However, our approach focuses on extracting common forgery traces from various methods, achieving generalization within a limited feature space without relying heavily on data.

2.2. Facial landmarks

Facial landmarks refer to the key points on the face, including the mouth, nose, eyes, and so on. In face-related tasks, facial landmarks have always been an important component. With the development of deep learning technology, researchers have achieved great results by modeling facial landmarks with deep neural networks.

In the early days, traditional image processing techniques and manually designed features were used to implement facial landmarks detection. With the development of deep learning technology, the method based on deep neural networks gradually became mainstream. Among them, Baltrusaitis et al. [22] used convolutional neural networks to model facial features and extract facial landmarks, achieving good results. In recent years, research on facial landmarks has been constantly advancing. For example, Dong et al. [23] separated different style elements in model training and prediction, and recombined them to improve accuracy. Currently, open-source toolkits such as Dlib [24] are widely used by researchers. In this work, we also employed this method to extract facial landmarks.

For the task of face forgery detection, some researchers have attempted to introduce face forgery data. Yang et al. [6] utilized facial landmark estimation and head pose error to determine whether the face has been tampered with. Their method utilized facial landmark data in low-level features without mapping them to a high-dimensional feature space. Subsequently, Sun et al. [25] used facial landmarks and differential facial landmarks as inputs to a two-stream network, but only selected facial landmark data as a single modality for the network input, abandoning color texture information. In our work, we combined facial landmarks with RGB modality and used our TTNet to lock the model's attention to the high-risk areas of facial features and contour regions where forgery traces are more likely to appear, obtaining more generalized feature representations, which can boost the generalization ability of TTNet.

3. Trail tracing network

In this section, we first elaborate on the motivation and give a brief overview of the proposed Trail Tracing Network (TTNet) in Section 3.1, then present the key components of TTNet in Section 3.2, and finally provide the loss function in Section 3.3.

3.1. Motivation and overview

A classifier with good generalization ability should classify the forgery data based on the authenticity of the face rather than other features of the face. Additional redundant information may hinder network generalization. We show a simulation example in Fig. 3. The left describes an over-learned classification where the real and fake images are not equally treated. The right side shows an excellent classification where forged images are treated equally regardless of identity, gender, etc. The fundamental reason for this issue is that the classification network on the right only focuses on the strongly generalizable forgery clues rather than other features. We will testify the simulation results in the subsequent experimental results.

Since most face forgery algorithms [5,26] involve replacing facial features or the entire face, strong generalizable clues can typically be found in a specific space. In order to enable the model to learn these clues, we propose to impose restrictions on the rich feature space extracted by the CNN, where RGB color texture is guided by the facial landmarks. This approach assists the model in capturing generalized forged traces in the face and reduces the impact of redundant information.

The architecture of the proposed TTNet is shown in Fig. 4. In the input stage, a facial landmark extractor is first employed to extract facial landmark data from the input face image. We denote the extracted facial landmark data by $X^{landmark} \in R^{N\times 68\times 2}$, where N is the batch size of the input, 68 is the number of interest points of facial landmarks, and 2 is the spatial coordinate dimension. After the dimension reduction by a fully connected operation, $X^{l_0} \in R^{N\times 68}$ is obtained for subsequent combination with the image information. In each convolution stage, the feature maps obtained from the input image are fully guided by the facial landmark embedding X^{l_i} (i represents the feature obtained after the ith convolutional stage) through Trace Enhancement Module (TEM). In this process, the two components Generalization Enhancement Module (GEM) and Redundancy Elimination Module (REM) of TEM are developed to implement the channel mapping for X^{l_i} ($i = 1, 2, 3, 4$), integrating both the facial landmarks and the feature maps, and the reduction of redundancy in feature maps. Finally, a fully connected operation is adopted to determine whether the image is forged or not.

3.2. Trace enhancement module

Different from previous works that used facial landmark data in shallow networks [6] or only utilized landmark data as the network's input [25], our model employs facial landmark data as auxiliary guidance for the color modality. To this end, we design a Trace Enhancement Module (TEM), which employs facial landmark data to enhance the facial forgery traces and capture generalizable forgery clues in the

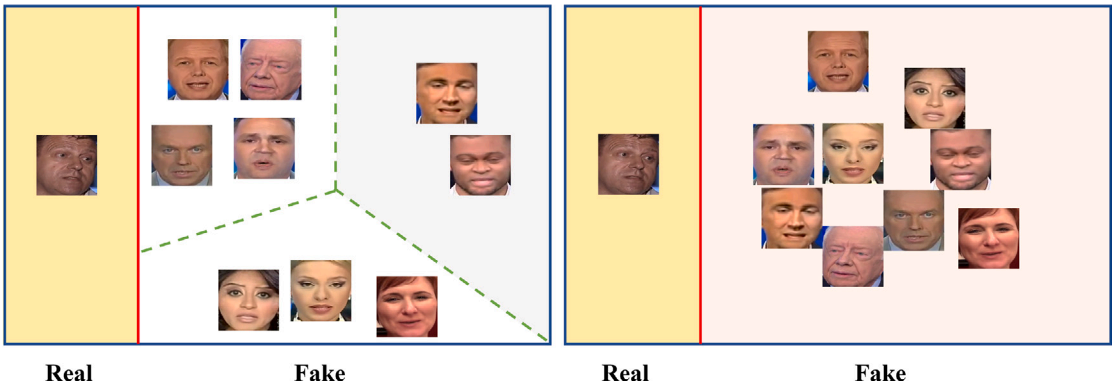

Fig. 3. A simulation example of the generalizable classification. The image contains one real face and nine fake ones. The left shows an over-refined classification where the classifier not only recognizes the authenticity of the face but also clusters the forged faces into different groups (according to identity, gender, etc.). The right presents a classification that only classifies the authenticity of the face without making any additional distinctions for the forged faces.

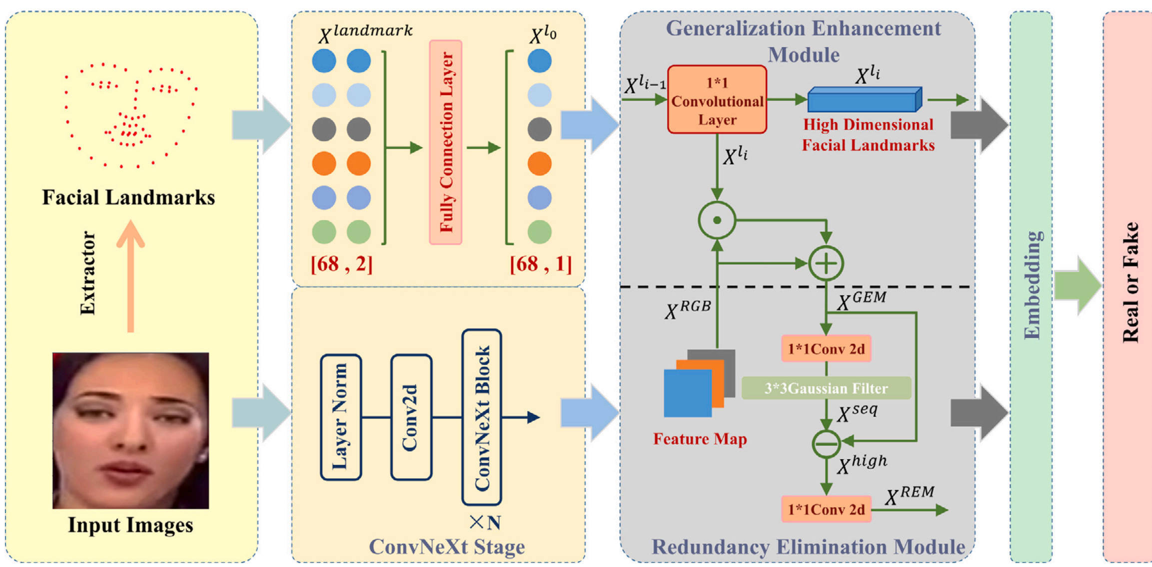

Fig. 4. The architecture of the proposed TTNet. In the input stage, facial landmark data is first extracted using a facial landmark extractor, and the dimension of the facial landmark data is transformed for subsequent fusion with the image. The image is then fed into a CNN, and at the end of each convolutional stage, the facial landmark data mapping by GEM is utilized to guide the feature embedding, which is then refined by REM. Finally, a fully connected layer is used at the end of the CNN to determine the authenticity of the image.

feature space. Specifically, TEM consists of two sub-modules, the Generalization Enhancement Module (GEM) and the Redundancy Elimination Module (REM). GEM aims to map facial landmark data to the channel dimension of feature maps and integrate them with RGB feature maps to capture the generalizable forgery traces around facial landmarks. REM targets to further refine the feature maps obtained from GEM, eliminate the impact of redundant information, and generate the refined embedding to the next convolutional stage. We will elaborate on the details in the following parts.

Generalization enhancement module. Assume $X^{RGB} \in R^{N\times C\times H\times W}$ denote the feature map obtained in the ith CNN convolution stage, where N is the batch size, C is the number of channels, and H and W are the dimensions of the feature map. Additionally, we have $X^{l_{i-1}} \in R^{N\times c}$, where N is the batch size and c is the number of channels obtained by the convolution operation of the previous stage. We use a 1×1 convolution operation D_{map} to align X^{l_i} and X^{RGB} in the channel dimension, and the operation is formulated in Eq. (1).

$$X^{l_i} = D_{map}(X^{l_{i-1}}, C), i = 0, 1, 2, 3, 4 \tag{1}$$

In the D_{map} operation, C denotes the target number of channels after convolution. At this point, X^{l_i} and X^{RGB} are aligned in terms of their channel dimensions. Similar to the approach in SENet [27], we use X^{l_i} as channel-level attention and multiply it with the feature map. It has been confirmed in [28] that each channel of the feature map responds to different spatial information. Therefore, the feature map at this stage will respond to different regions of the face, and the source of these regions is the affine transformation of several facial landmarks. The specific operation is shown as follows.

$$X^{GEM} = X^{l_i} \cdot X^{RGB} + X^{RGB} \tag{2}$$

In this operation, X^{GEM} is the feature map output that combines facial landmark data. We have achieved constraint on the feature space in this stage, where the constraint is the facial landmark data. Afterward, X^{GEM} will be employed as the input for the next module.

Redundancy elimination module. After the GEM stage, we initially constrained the feature map with facial landmarks. However, due to the existence of noise and the influence of background information, attention to facial data may be directed to these areas, which is contrary to our model mechanism. In order to reduce the impact of this redundant information, we need to refine the feature map further. Assuming that we obtain $X^{GEM} \in R^{N\times C\times H\times W}$ after GEM, we need to use a 1×1 convolution operation to reduce the channel dimension from C to 1.

The purpose of this operation is to compress various attention information of the feature map into one channel for subsequent filtering. The specific operation is shown as follows:

$$X^{seq} = D_{sq}(X^{GEM}, 1) \tag{3}$$

where D_{sq} is the 1×1 convolution operation used to compress the channel dimension, and 1 is the target channel dimension size. X^{seq} is the feature map with compressed channel dimensions, which merges the corresponding areas separated by the channel dimension in the previous feature map into one image to achieve information integration. Then, we use a Gaussian filter to reduce the amplitude of high-frequency components by blurring the feature map. And we use the operation of difference to achieve the separation of the image frequency domain and retain the high-frequency components of the image in the subsequent operation. The detailed procedure is illustrated in Eq. (4).

$$X^{high} = X^{seq} - gaussian(X^{seq}) \tag{4}$$

At this point, X^{high} obtained by frequency domain separation on X^{seq} has eliminated the influence of noise and avoided the decline in the effect of facial landmarks in subsequent operations. It is worth noting that we do not choose DCT or SRM operations commonly used in previous frequency domain-related works [29] to separate image frequency components to reduce computational costs.

Finally, we adopt the 1×1 convolution operation D_{exp} to expand the feature map back to the original size for further computation in the neural network. The particular process is described as follows:

$$X^{REM} = D_{exp}(X^{high}, C) \tag{5}$$

where C is the expanded feature dimension. At this point, the output X^{REM} has the same size as the output X^{RGB} in the convolution stage, which has no impact on the shape of the subsequent input to the network. This is also the advantage of the embedding module we design.

3.3. Loss function

The cross-entropy loss is widely used in deepfake detection tasks, as it allows the model to learn to distinguish between real and forged images by minimizing the difference between the predicted and actual probability distributions. By minimizing the cross-entropy loss, the model can learn to assign high probabilities to real images and low probabilities to fake images, thereby improving its performance in detecting deepfake images. Therefore, in our work, we adopt the same choice as previous researchers and utilize cross-entropy loss as our loss function. The specific formulation is described in Eq. (6).

$$\mathcal{L} = -\frac{1}{N}\sum_{j=1}^{N}(y_j log(\hat{y}_j) + (1 - y_j) log(1 - \hat{y}_j)) \tag{6}$$

where N is the number of samples, y_j is the ground-truth label (0 or 1) of the jth sample, and $\hat{y}_j$ represents the predicted probability of the model for the jth sample. Finally, we present our end-to-end algorithm in Algorithm 1.

Algorithm 1 TTNet: Trail Tracing Network

Input: A training dataset S with max_iterations m, num_batch n, learning rate α
Output: A deepfake detection model with generalization ability
Initialization: $\theta^0, l = 0$

```
1:  for e = 1 to m do
2:    for b = 1 to n do
3:      Sample a batch S^b from S
4:      Extract Facial Landmarks and image feature X^L, X^RGB
5:      Compute X^GEM based on Eq. (2)
6:      Compress the channel dimension using Eq. (3)
7:      Compute X^high based on Eq. (4)
8:      Expand the feature map using Eq. (5)
9:      Compute cross-entropy loss L for (6)
10:     Update θ : θ^{l+1} ← θ^l − α·Adam(θ^l, L)
11:     l ← l + 1
12:   end for
13: end for
```

4. Experiment

In this section, we first introduce our experimental settings and datasets used in our work in Section 4.1. Then, in Section 4.2, in-dataset evaluation and cross-dataset evaluation are conducted and we analyze the experimental results against state-of-the-art methods. In Section 4.3, we execute ablation experiments to study the effectiveness of each component in our model.

4.1. Experimental setting

Experimental implementation. In the data extraction stage, we utilize S3FD [30] to extract 32 facial images evenly from each video and resize them to the size of $224 * 224$. After obtaining the facial images, we employ the Dlib [24] software package to extract 68 facial landmarks for each image. In the training stage, we adopt the ConvNeXt-B [31] model pre-trained on ImageNet [32] as the backbone network for our TTNet. We use the AdamW optimizer with moment betas of (0.9, 0.999) and a learning rate of $4 * 10^{-5}$ and employ a cosine learning rate schedule to iterate our learning rate, with a period of 100. The batch size is set to 24 and we train our model for 500 epochs. Additionally, we employ the cross-entropy loss function as our loss function. To compare with state-of-the-art works, we report the detection performance by the video-level AUC. The proposed method is implemented based on Pytorch and trained on two NVIDIA RTX 3090 GPUs.

Datasets. We evaluated the detection performance and generalization capability of TTNet on multiple datasets, including FaceForensics++ (FF++) [5], DFDCp [26], Celeb-DF [33], and WildDeepfake [34].

- **FaceForensics**++ is a video dataset for facial manipulation detection, which consists of 1,000 original videos and 1,000 manipulated videos artificially synthesized using four commonly used facial manipulation techniques, namely, Deepfakes, Face2Face, NeuralTextures, and FaceSwap. The purpose of this dataset is to facilitate research and development of facial manipulation detection techniques, enhance public awareness of facial manipulation, and prevent its misuse.
- **DFDCp** contains 1,140 real and 4074 manipulated videos and is regarded as one of the most challenging datasets due to the high quality of the manipulated videos, which are difficult to distinguish from real ones. The manipulated videos were generated using state-of-the-art facial manipulation techniques, resulting in subtle manipulation artifacts that make it hard to detect. Therefore, DFDCp is a valuable resource for advancing research on deepfake detection and improving public awareness of media authenticity and manipulation.
- **Celeb-DF** consists of 590 original videos collected from YouTube and 5,639 corresponding high-quality manipulated videos generated using deepfake techniques. The manipulated videos were synthesized using deepfake facial manipulation technology and are of high quality. The uniqueness of this dataset is that it covers multiple celebrity interview videos, so the videos are of high quality with relatively fixed background information.

Table 1
In-dataset evaluation on FF++ (HQ). We utilize four manipulation methods, including deepfakes (DF), face2face (F2F), faceswap (FS), and neuraltextures (NT) as our training data, and test our model on all four manipulation methods. The detection performance is evaluated by the video-level AUC.

Training set	Methods	Intra-dataset(AUC(%))				
		DF	F2F	FS	NT	Avg
DF	Xception [35]	99.9	71.8	42.1	87.7	75.4
	FTCN [36]	99.3	76.0	53.5	87.4	79.1
	LipForensics [37]	99.7	75.7	36.6	90.8	75.7
	ADAL [38]	99.6	79.4	**73.6**	81.9	83.6
	Mre-Net [39]	99.9	87.4	49.1	81.0	79.3
	CDIN [40]	99.8	79.9	54.2	88.1	80.5
	ConvNeXt-B [31]	99.8	78.2	48.6	90.1	79.1
	TTNet(Ours)	**99.9**	**87.6**	58.2	**92.3**	**84.5**
F2F	Xception [35]	51.4	99.3	26.6	46.3	55.9
	FTCN [36]	81.7	98.9	76.4	85.9	85.7
	LipForensics [37]	88.4	99.2	72.3	91.8	87.9
	ADAL [38]	90.3	99.1	69.4	73.1	82.9
	Mre-Net [39]	**91.2**	99.9	52.1	83.2	81.6
	CDIN [40]	84.5	99.6	70.5	91.9	86.6
	ConvNeXt-B [31]	84.2	99.0	70.9	88.2	85.5
	TTNet(Ours)	89.7	**99.9**	**76.4**	**92.9**	**89.7**
FS	Xception [35]	72.2	56.6	99.9	75.6	76.1
	FTCN [36]	**88.1**	69.7	98.9	76.7	83.4
	LipForensics [37]	54.0	68.6	99.5	38.4	65.1
	ADAL [38]	72.3	70.2	99.9	62.1	76.2
	Mre-Net [39]	73.3	63.5	99.9	55.1	72.9
	CDIN [40]	78.1	67.7	99.8	77.2	80.7
	ConvNeXt-B [31]	74.2	66.7	99.6	74.2	78.6
	TTNet(Ours)	84.5	**70.5**	**99.9**	**79.8**	**83.6**
NT	Xception [35]	84.5	77.4	39.1	98.7	74.9
	FTCN [36]	90.0	88.1	62.4	98.0	84.6
	LipForensics [37]	**95.5**	89.0	52.0	96.5	83.3
	ADAL [38]	90.9	63.2	**78.4**	99.2	82.9
	Mre-Net [39]	96.7	49.9	85.3	99.2	82.7
	CDIN [40]	92.9	90.1	62.2	99.8	86.3
	ConvNeXt-B [31]	92.0	88.5	60.4	99.7	85.1
	TTNet(Ours)	94.0	**90.5**	66.1	**99.9**	**87.6**

- **WildDeepfake** is a dataset from the real world, which contains 3805 real clips and 3509 fake clips. Due to the changeable background information and shooting angles of the video, its content is closer to the real-world situation, which has always caused this dataset to be full of challenges for the detection model.

4.2. Experimental analysis

In-dataset evaluation. To demonstrate the in-dataset detection capability of our method, we conduct extensive intra-class experiments and compare our results with previous state-of-the-art works. We train our model on different manipulation methods of the FaceForensics++ (FF++) dataset and test it on all manipulation methods of FF++. It is worth noting that the FF++ dataset consists of three categories of data, including raw data (RAW), high-quality compressed videos (HQ), and low-quality compressed videos (LQ). We employ the HQ data commonly used in previous works as our training data and conduct comparisons against state-of-the-art detection methods as well as our backbone network. The results are reported in Table 1. The experimental results clearly indicate that TTNet outperforms existing methods in the majority of cases (with average AUC values surpassing the previous best by 0.9%, 1.8%, 2.9%, and 1.3% across four different training sets), which demonstrates that the features extracted by TTNet are discriminative for the in-dataset evaluation.

Cross-dataset evaluation. If we train and test a model on one dataset with the same distribution, the excellent detection results can only prove that the method sufficiently fits the forgery traces generated by the forgery techniques used in that dataset. However, an outstanding detector should be able to perceive forgery traces left by multiple forgery methods, i.e., possess strong generalization capabilities. In the proposed TTNet, by incorporating facial landmarks as auxiliary data along with the image and the design of TEM to enhance the suspicious facial forgery trace regions, the model can learn to obtain highly generalized feature representations. To verify this, we design the cross-dataset evaluation on several benchmarks to expand diversity, and the results are reflected in Table 2. We adopt FF++(HQ) as the training data and test on Celeb-DF, DFDCp, and WildDeepfake datasets. Since the distribution of FF++ is different from these three datasets, the model's detection performance would experience some decline during cross-dataset evaluation. However, our method achieves the best generalization results among all the detectors (TTNet outperforms the previous best methods by an average of 0.8%), which proves that our approach possesses excellent generalization capabilities.

Table 2
In the cross-dataset evaluation, we train on the complete FF++(HQ) dataset and then test on three datasets, namely Celeb-DF, DFDCp, and WildDeepfake, using video-level AUC as the performance evaluation metric.

Training set	Methods	Test set (AUC(%))			
		Celeb-DF	DFDCp	WildDeepfake	Avg
FF++	Xception [35]	88.0	79.2	58.2	75.1
	MAT [41]	72.5	70.9	65.9	69.7
	Lipforensics [37]	82.4	73.5	68.2	74.7
	Face-x-ray [15]	79.5	65.5	64.7	69.9
	ADAL [38]	84.6	78.5	67.0	76.7
	CADDM [42]	93.8	73.8	69.1	78.9
	LAA-Net [43]	**94.4**	86.9	67.1	82.8
	ConvNeXt-B	90.9	78.2	65.4	78.1
	TTNet (Ours)	94.0	**87.5**	**69.5**	**83.6**

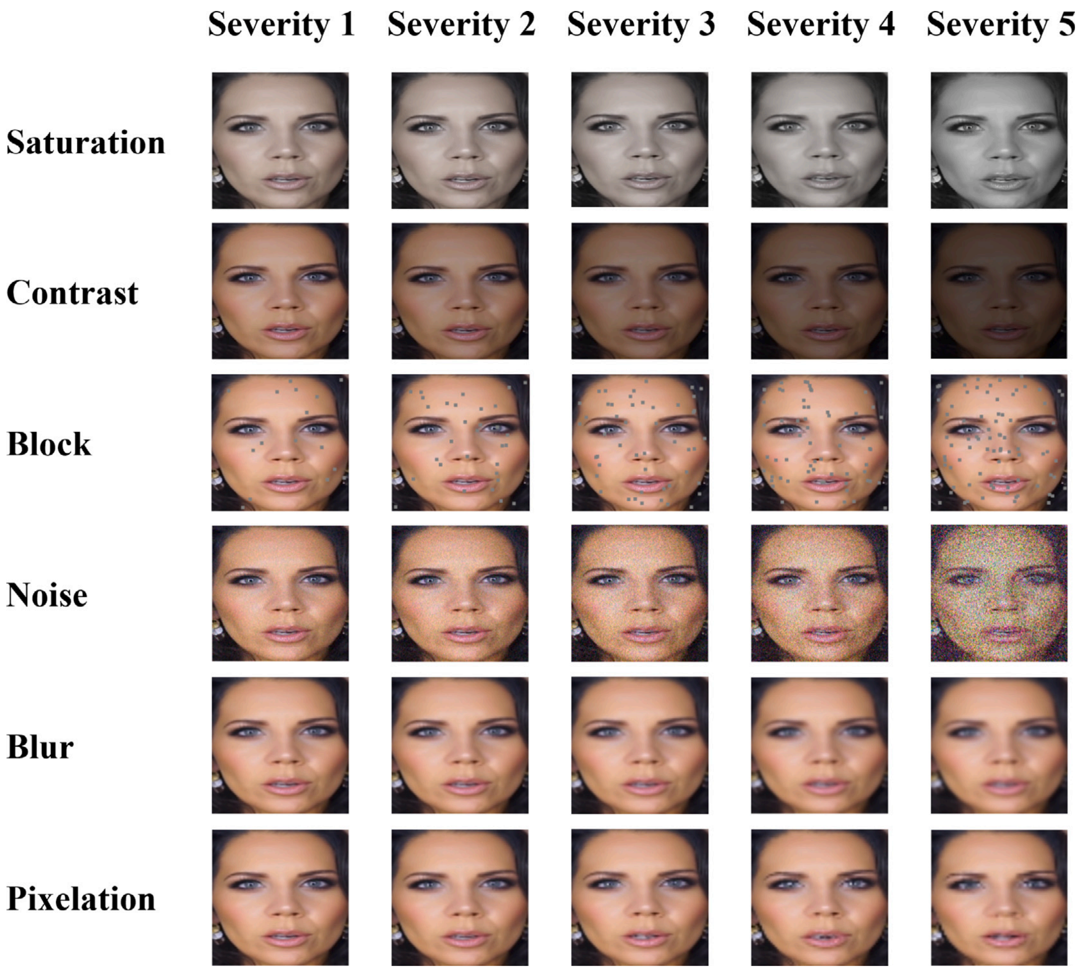

Fig. 5. Six types of perturbations. Compared with the original, it can be observed that each type of perturbation can cause varying degrees of damage to the image, thus negatively affecting the detection performance of the model.

Distortion evaluation. Due to the fact that most of the current forged images come from social media, images may experience various degrees of distortion during transportation over the Internet. Our network should not only achieve ideal results on standard datasets but also overcome various obstacles in practical applications. Therefore, we design the perturbation experiments based on experimental settings in Lipforensics [37]. Specifically, we select Color Saturation, Color Contrast, Local Block-wise, Gaussian Noise Blur, Gaussian Blur, and Pixelation as perturbation methods (the perturbation effects are shown in Fig. 5), and execute experiments at five levels of perturbation. The results are shown in Table 3, where our method achieves the best average detection performance among all detectors. In Local Block-wise distortion, our method is slightly inferior to the state-of-the-art method LAA-Net. This is because block occlusion affects the integration of facial landmarks with RGB data. The fact that it does not lead to a significant performance drop also indicates the robustness of our method.

Visualization. In order to better demonstrate the generalization ability of our proposed method and the robustness of our classification, we design a visualization scheme. We train TTNet and our backbone network on FF++(HQ) dataset and then apply t-SNE [47] visualization method to visualize the feature vectors from the last layer of the models. During the visualization stage, we employ 10 individuals with 32 images each, a total of 320 images to display the feature embedding. As shown in Fig. 6, results obtained by the backbone network (the left one) not only distinguish between real and fake images but also recognize images from different individuals. This indicates that the backbone network is overly influenced by forgery-unrelated features, such as skin tone and background, leading to excessive fine-grained classification. While such fine-grained features may improve detection performance within the dataset, they may lead to model overfitting, causing the model to exhibit poor generalization when encountering unseen forgery methods. In contrast, the visualization results of TTNet on the right side of the figure show that the model does not make further distinction between different forgery methods or individuals, but only distinguishes between real and fake images. This demonstrates that our method can identify forgery clues that commonly exist across various forgery methods, allowing it to better capture the fundamental differences between real and fake images, thereby demonstrating superior generalization ability.

In addition, to better demonstrate the embedding encoding of our network, we use CAM [48] to locate which regions are activated for detecting forgery. The visualization results in the second row of Fig. 7 show that in some cases, the backbone network focuses on irrelevant forgery regions, such as the background or inner facial areas. Unlike the backbone network, our network, with the guidance of facial landmarks and the design of TEM, pays more attention to the common forgery clues near the facial landmarks and concentrates on forgery points after mapping through multiple convolutional stages. The specific effect is shown in the third row of Fig. 7, where for the five types of forgery methods in the figure, the attention region of the model with the guidance of facial landmarks is significantly different from the one without, and the attention points are also consistent with our analysis. This visualization further indicates that facial landmarks encourage the network to capture more general forgery features.

Table 3
Distortion evaluations on FF++. We train the model on the FF++ (RAW) dataset and test on data with six types of perturbations, each with 5 levels. The table shows the average result among five levels of perturbation and the performance is evaluated by the video-level AUC.

Method	Original	Saturation	Contrast	Block-wise	Noise	Blur	Pixelation	Avg
Xception [35]	99.8	99.3	98.6	99.7	53.8	60.2	74.2	80.9
CNN-aug [44]	99.8	99.3	99.1	95.2	54.7	76.5	91.2	86.0
PatchForensics [45]	99.8	84.3	74.2	99.2	50.0	54.4	56.7	69.8
Face-x-ray [15]	99.9	97.6	88.5	99.1	49.8	63.8	88.6	55.2
CNN-GRU [46]	99.8	99.0	98.8	97.9	47.9	71.5	86.5	83.6
LipForensics [37]	99.9	99.9	99.6	87.4	73.8	96.1	95.6	92.0
ITSNet [19]	99.9	99.9	99.3	97.2	81.9	98.1	96.9	95.5
CADDM [42]	99.9	99.6	99.8	99.8	87.4	99.0	98.8	97.4
LAA-Net [43]	99.9	99.9	**99.9**	**99.9**	53.9	98.8	**99.8**	93.1
TTNet (Ours)	**99.9**	**99.9**	99.8	97.3	**89.5**	**99.4**	99.2	**97.5**

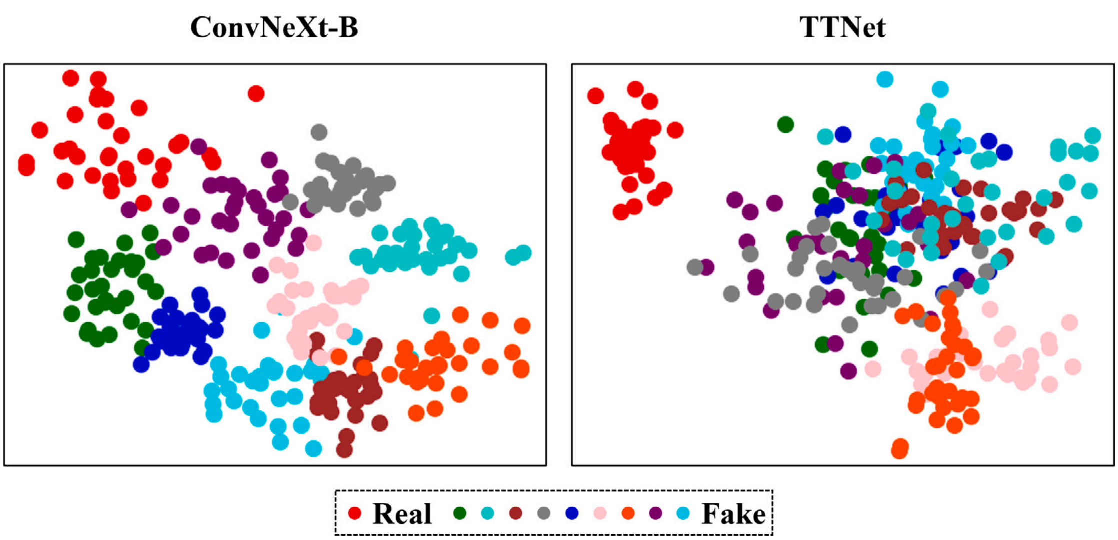

Fig. 6. Classification analysis. We select 1 real face and 9 fake faces, each with 32 images, to show the classification results by the backbone network (left) and the proposed TTNet (right) on the test set. The backbone network (left) not only distinguishes between real and fake images but also recognizes images from different individuals. In contrast, TTNet only distinguishes between genuine and fake images, which shows a better generalization ability.

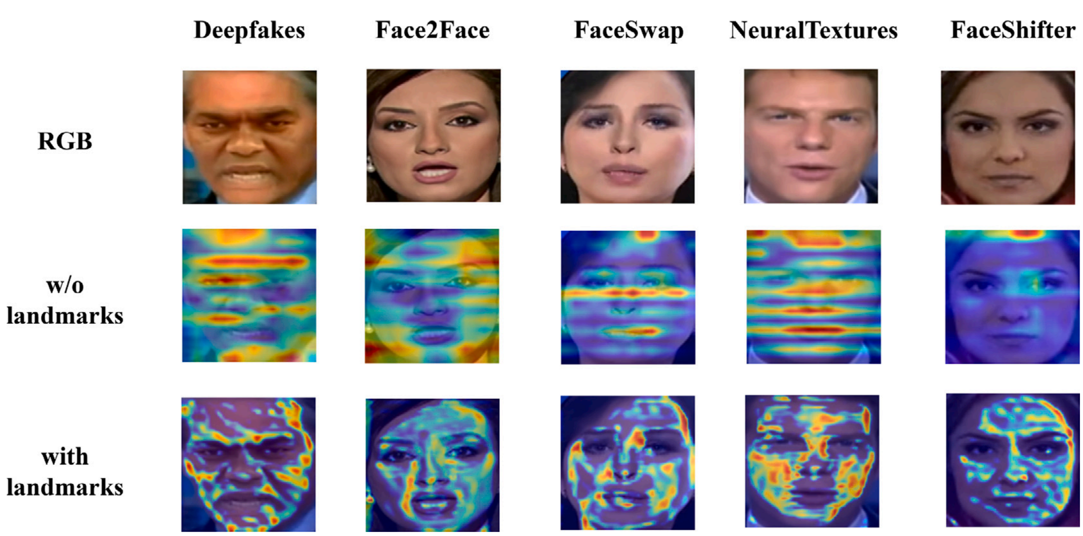

Fig. 7. Visualization by CAM. It is evident that introducing facial landmarks has a significant impact on the model's attention.

Table 4
With the addition of more TEMs at various stages of the backbone network, the detection and generalization capabilities of the model gradually increase, demonstrating the effectiveness of the modules we designed.

Training set	Stages				Test set(AUC(%))			
	1	2	3	4	FF++	Celeb-DF	DFDCp	WildDeepfake
FF++	–	–	–	–	95.5	90.9	78.2	65.4
	✓	–	–	–	96.2	91.5	78.8	68.3
	✓	✓	–	–	98.3	93.4	79.2	68.9
	✓	✓	✓	–	99.4	93.8	79.7	69.1
	✓	✓	✓	✓	**99.7**	**94.0**	**87.5**	**69.5**

Table 5
By adding the GEM and REM modules separately to the backbone network, the model's detection ability is enhanced, validating the effectiveness of GEM and REM.

Training set	Components		Test set(AUC(%))			
	GEM	REM	FF++	Celeb-DF	DFDCp	WildDeepfake
FF++	–	–	95.5	90.9	78.2	65.4
	✓	–	98.4	92.2	83.7	68.9
	–	✓	97.7	91.9	81.5	68.3
	✓	✓	**99.7**	**94.0**	**87.5**	**69.5**

4.3. Ablation studies

Effectiveness of components. To demonstrate the effectiveness of TEM we design, we conduct experiments by inserting TEM at the end of the four convolution stages of ConvNeXt-B. FF++(HQ) is employed as the training data, and FF++(HQ), Celeb-DF, DFDCp, and WildDeepfake are used as the test data. The experimental results are reported in Table 4, where the detection and generalization capabilities of the model gradually increase with the gradual addition of TEM, and the best performance is achieved when TEM is inserted into all convolution stages of TTNet. In addition, to separately validate the effectiveness of the components GEM and REM in TEM, we conducted experiments by inserting GEM and REM into the backbone network. The experimental results, as shown in Table 5, indicate that both adding GEM and REM improve the detection performance. This demonstrates the effectiveness of the GEM and REM module. Furthermore, it can be observed that the addition of GEM yields better results than REM. This may be because REM primarily removes low-frequency background information, but lacks the feature enhancement capabilities of GEM, which limits its improvement in forgery detection. In summary, this proves the effectiveness of TEM.

Effectiveness of facial landmarks. Similarly, to investigate the impact of facial landmarks, we conduct similar experiments. As the facial landmark data we employ have 68 dimensions, we design variants of 17, 34, and 51 dimensions using sparse sampling to demonstrate the effectiveness of all variants. We also employ FF++(HQ) as the training data, and FF++(HQ), Celeb-DF, DFDCp, and WildDeepfake are used as the test data. The results are reflected in Table 6, where the detection and generalization capabilities are the most inferior when using the sparsest input of 17 dimensions. As the facial landmark data becomes denser, we can obtain better detection and generalization capabilities. The best performance is achieved when using all 68 dimensions of facial landmark data. This demonstrates the effectiveness of facial landmark data.

5. Conclusion

In this paper, we investigated the generalization of face forgery detection methods and found that the key was to identify the common forgery traces left by various forgery methods. Based on this observation, we proposed our TTNet to capture these clues, where we designed a novel trace enhancement module to combine facial landmarks and RGB data. With the guidance of facial landmarks, forgery clues of a specific spatial range can be captured to yield a generalized feature representation. To verify the effectiveness of our proposed method, we conducted experiments on multiple datasets. Compared with previous state-of-the-art works, our method achieved the best results on multiple evaluation metrics, demonstrating its good detection and generalization capabilities. To verify the effectiveness of the proposed components, we also conducted ablation experiments to demonstrate the positive significance of our innovation from multiple perspectives. At the same time, we notice that the effectiveness of TTNet can be limited by the accuracy of facial landmark extraction. If the landmark extraction is affected by occlusion, it may impact the detection performance. In the future, we will focus on addressing this issue. Finally, we believe that our approach can be applied to various real-world scenarios and contribute to the fight against deepfake technology.

CRediT authorship contribution statement

Qi Gao: Writing – review & editing, Writing – original draft, Visualization, Investigation, Data curation. **Baopeng Zhang:** Writing – review & editing, Supervision, Project administration, Data curation. **Jianghao Wu:** Writing – original draft, Visualization, Formal analysis, Conceptualization. **Wenxin Luo:** Visualization, Validation, Software, Investigation. **Zhu Teng:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Jianping Fan:** Resources, Project administration, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Table 6
As the dimensionality of the facial landmark features increases, the detection and generalization capabilities of the model gradually increase, and the best performance is achieved when using the original data with 68 dimensions.

Variant input	Training set	Test set(AUC(%))			
		FF++	Celeb-DF	DFDCp	WildDeepfake
17-dimension	FF++	98.4	92.6	78.3	67.9
34-dimension		98.9	93.1	79.5	68.2
51-dimension		99.5	93.7	80.0	68.9
68-dimension (TTNet)		**99.7**	**94.0**	**87.5**	**69.5**

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